Chapter 2
Supervised Learning

Course Outline

Linear Models and Regularization

- Linear Regression
- Logistic Regression
- Regularization Techniques

Decision Trees and Ensemble Methods

- Decision Trees
- Bagging: Random Forests
- Boosting
- Stacking and Blending

Advanced SVM and Optimization

- Introduction to Support Vector Machines (SVM)
- Kernel trick: linear, polynomial, RBF, sigmoid
- Optimization techniques: SGD, Adam

Introduction

- **Supervised learning** constitutes the **fundamental cornerstone of modern machine learning**.
- It encompasses the set of methods that enable a computer system **to learn to predict an output (label)** from a set of **input–output** examples known in advance.

Introduction

- **Supervised learning** makes it possible, from a training sample $D_m = \{(x_1, y_1$ [illegible] en x and y.
  - x_i = char
  - y_i = labe
  - An elem
- The centra

The model learns from the **past** (known data) in order to **generalize** its predictions to **new data**.

$$f : X \rightarrow Y$$

  which, given a vector of features X, best predicts the target value Y.
- We call *f(x)* a **model** or a hypothesis.

Introduction

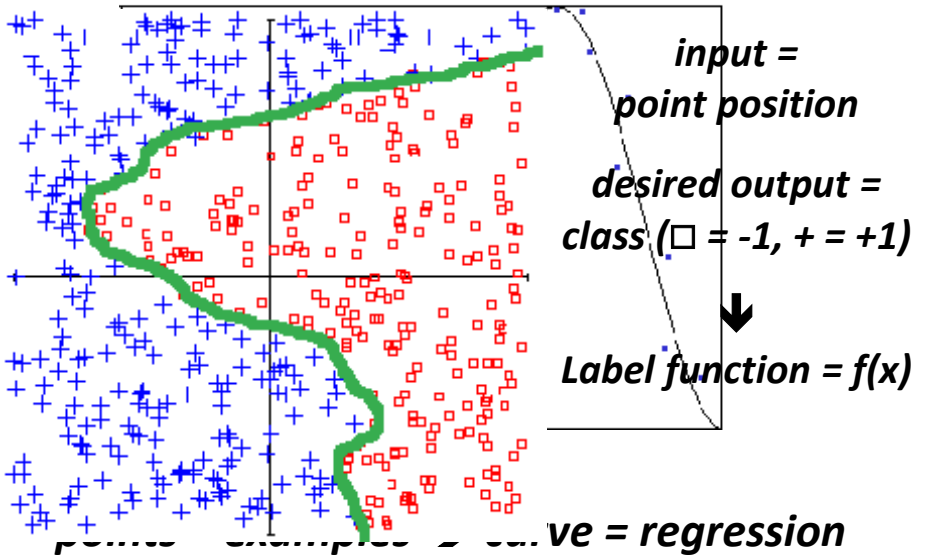
- A **learning algorithm** receives:
  - A **training set** containing pairs (x_i, y),
  - A **prediction objective**, often expressed as a **cost function** to be minimized,
- And returns a **trained model** capable of predicting the output for new inputs.

Introduction

- **Types of supervised learning**
- The central idea is to learn the function $f: X \rightarrow Y$
- Problems are classified according to the type of output domain:
  - If Y = $\mathbb{R}$, this is referred to as regression.
  - If Y is a finite discrete set, this is referred to as classification.
    - If Y has exactly two elements, this is referred to as binary classification.

Introduction

- **Types of supervised learning**



Task type	Description	Examples
Regression	Predict a **continuous value**	Predicting the price of a house, the temperature, a salary, a student's grade…
Classification	Predict a **category** or **label**	Spam detection, medical diagnosis, image recognition…

Linear Models and Regularization

1.1. Linear Regression

Period	Scientist	Contribution
1805	**Adrien-Marie Legendre**	Introduced the **least-squares method**, making it possible to fit a straight line to experimental data.
1809	**Carl Friedrich Gauss**	Mathematically formalized the method and demonstrated that it minimizes the sum of squared errors.
1886	**Francis Galton**	Introduced the term "**regression**" in his study of the relationship between **the heights of parents and those of their children.**
1901–1913	**Karl Pearson**	Extended regression to several variables and introduced the notion of **linear correlation**.

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Linear Models and Regularization

1.1. Linear Regression

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1829	**Ca** **G**	
1886	**Francis Galton**	**[illegible] the term "regression" in his study of the relationship between the heights of parents and those of their children.**
1901–1913	**Karl Pearson**	Extended regression to several variables and introduced the notion of **linear correlation**.

- The word "regression" is said to have been introduced by Galton in the 19th century.
- By modeling individuals' heights as a function of their fathers' heights, Galton observed a return (regression in English) toward the average height.
- Taller-than-average fathers tend to have children shorter than themselves, and vice versa for shorter-than-average fathers.

Linear Models and Regularization

1.1. Linear Regression

- Linear regression is a **supervised learning** algorithm used to predict a **continuous target variable**, for example:
  - the price of a house,
  - the temperature,
  - a test score,
  - a taxi travel time,
  - the number of daily visitors to a website
  - a company's revenue for the coming year.
- Its objective is to model the relationship between
  - a **continuous target variable** y
  - and one or more **explanatory variables** $x_1, x_2, \ldots, x_n$.

Linear Models and Regularization
1.1. Linear Regression

- **Mathematical formulation**

- The model assumes a linear relationship between the input features X (explanatory variables) and the target variable Y (dependent variable).
- **Single-variable model (Simple Linear Regression):**

$$y_i = w_0 + w_1 x_i$$

Linear Models and Regularization

1.1. Linear Regression

- **Mathematical formulation**

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- **Single-variable model (Simple Linear Regression):**

$$y_i = w_0 + w_1 x_i$$

- **Multi-variable model (Multiple Linear Regression):**

$$y_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \cdots + w_n x_{in} + \varepsilon_i$$

- where:
  - y_i :observed actual value,
  - x_{ij} :j-th input variable for example i,
  - w_j :coefficients (weights) of the model,
  - ε_i :residual error.

In vector form:

$$\hat{y} = Xw$$

Linear Models and Regularization

1.1. Linear Regression

- **Cost Function (Loss Function)**
- To train the model, we must determine the weights w_i that minimize the error between the predictions $\hat{y}$ and the actual values y.
- The most common cost function is the **Mean Squared Error (MSE)**, which measures the average of the squared errors:

$$J(w) = \frac{1}{2m} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2$$

  - m : number of examples
  - y_i : actual value
  - $\hat{y}_i$: predicted value

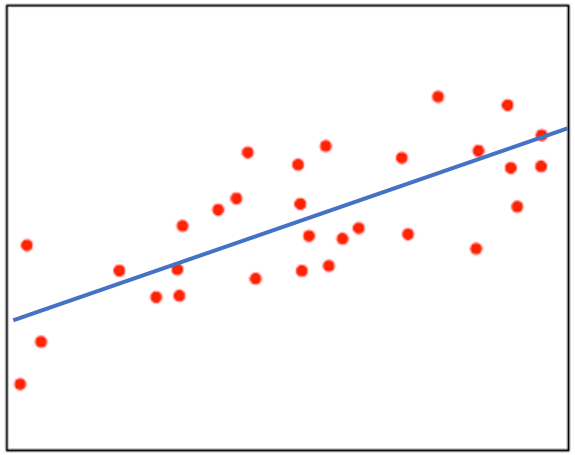
Linear Models and Regularization
1.1. Linear Regression

- **Cost Function (Loss Function): MSE**

- The objective of **learning** is
  - to find the parameter vector w that **minimizes** this error $J(w)$,
  - which amounts to finding the straight line (or hyperplane) that passes **as closely as possible** to all the data poir

x	y
0.86	2.49
0.09	0.83
-0.85	-0.25
0.87	3.10
-0.44	0.87
-0.43	0.02
-1.10	-0.12
0.40	1.81
-0.96	-0.83
0.17	0.43





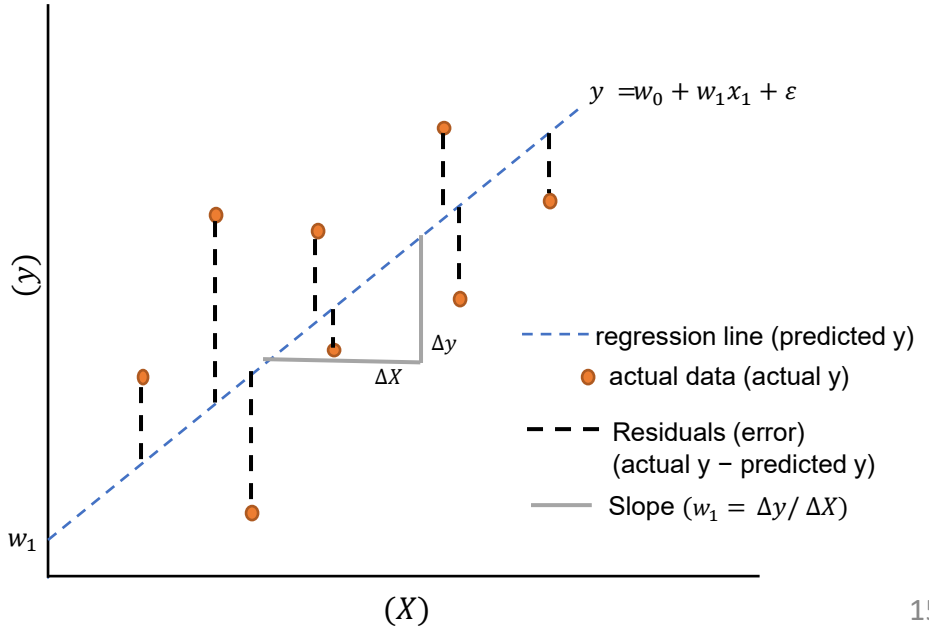
Linear Models and Regularization

1.1. Linear Regression

- **Cost Function (Loss Function): MSE**

- The objective of **learning** is
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  - which amounts to finding the straight line (or hyperplane) that passes **as closely as possible** to all the data points.

- **Y:** the value we want to predict.
- $\boldsymbol{w_1}$:the slope of the line.
- ***x:***our input value.
- $\boldsymbol{w_0}$:the intercept.



Linear Models and Regularization
1.1. Linear Regression

- **Cost Function (Loss Function): MSE**
- **Analytical Method (Normal Equation):**
  - Finds the optimal solution $\widehat{w}$ directly through a matrix formula, without iterations.
  - **Formula:** $\widehat{w} = (X^T X)^{-1} X^T y$
  - **Advantage**: exact solution, no iterations
  - **Limitation:** Becomes very slow when the number of features n is very large, since it requires inverting the matrix ($X^T X$).

Linear Models and Regularization
1.1. Linear Regression

- **Cost Function (Loss Function): MSE**

- **Iterative Method (Gradient Descent):**
  - A numerical optimization method that progressively adjusts the model weights in order to minimize the cost function $J(w)$.
  - Unlike the analytical method, it **does not compute the solution directly**, but **converges toward it through successive iterations**.
  - **Formula:** $w_j = w_j - \alpha \frac{\partial J(w)}{\delta w_j}$ where:
    - α is the **learning rate**, which determines the size of each step toward the minimum,
    - $\frac{\partial J(w)}{\partial w_j}$is the **gradient** (the local slope of the cost function).

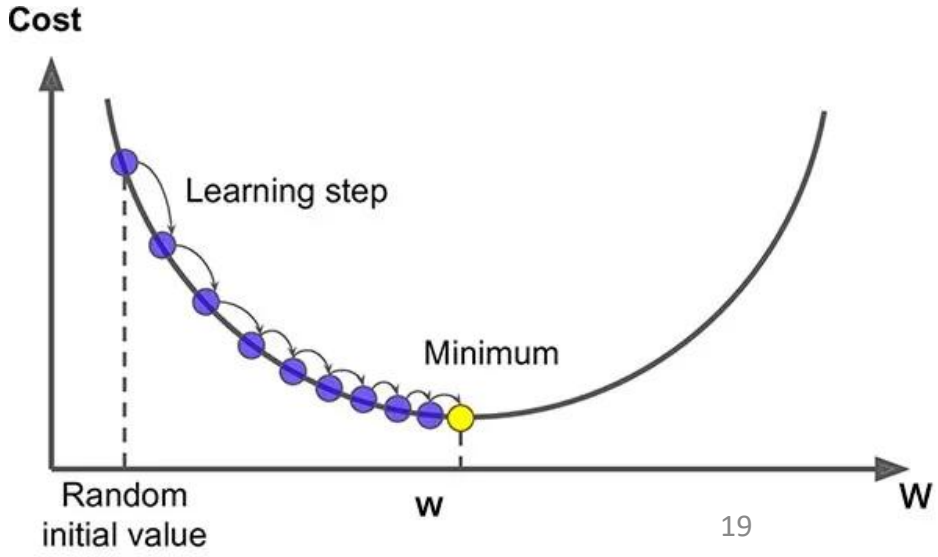
Linear Models and Regularization
1.1. Linear Regression

- **Cost Function (Loss Function): MSE**

- **Iterative Method (Gradient Descent):**
  - **General principle:**
  - At each iteration, the coefficients w_j are adjusted **in the direction opposite to the gradient**,
  - that is, the direction of **steepest decrease** of $J(w)$.
  - The algorithm stops when the changes become negligible (convergence).

Linear Models and Regularization
1.1. Linear Regression

- **Cost Function (Loss Function): MSE**
- **Iterative Method (Gradient Descent):**
- The learning rate is crucial for the convergence of the algorithm:
  - If α is too small, convergence will be very slow.
  - If α is too large, the algorithm may diverge or oscillate around the minimum.



Linear Models and Regularization
1.1. Linear Regression

- **Cost Function (Loss Function): MSE**
- **Iterative Method (Gradient Descent):**
  - **Advantage:**
  - Efficient for large datasets and high-dimensional models.
  - Does not require inverting the matrix $(X^T X)$.
  - Applicable to any model whose cost function is differentiable.

Linear Models and Regularization
1.1. Linear Regression

- **Cost Function (Loss Function): MSE**
- **Iterative Method (Gradient Descent):**
  - **Limitation:**
  - Does not yield **an exact solution** but an **approximation** that depends on the learning rate and the number of iterations.
  - May **converge slowly** if α is too small, or **diverge** if α is too large.
  - Sensitive to **data scaling** (often requires prior normalization).

Linear Models and Regularization

1.1. Linear Regression

- **Example: salary prediction**
- We have a dataset $(x, y)$where:
  - x= number of years of experience,
  - y= annual salary.
- Objective: estimate y from x.

$$y = w_0 + w_1 x + \varepsilon$$

- After training, we obtain for example:

$$\hat{y} = 40000 + 2500x$$

- Interpretation:
  - $w_0 = 40000$: base salary (with no experience),
  - $w_1 = 2500$: average increase of 2,500 DZD per additional year of experience.

Linear Models and Regularization
1.2. Logistic Regression

- **Linear regression** is used to predict a **continuous value**:

$$\hat{y} = w^T X + w_0$$

- In many machine learning settings, the **target variable** is not numerical:
  - Email → *spam / not spam*
  - Patient → *sick / healthy*
- Linear regression **is not suitable** for these problems:
  - It can **take values outside [0, 1]**, making them impossible to interpret as probabilities.
  - It does not **support classification**.

Linear Models and Regularization
1.2. Logistic Regression

- Logistic regression is a natural extension of linear regression.
- It retains the linear model but changes the output function.
- It can predict classes by estimating a membership probability.
- **Logistic regression** is a supervised model used to solve **binary classification problems**.
  - It predicts the **probability** that an observation belongs to a **given class**.
  - It constitutes the baseline model of **supervised classification**.

Linear Models and Regularization
1.2. Logistic Regression

- Logistic regression retains the **linear structure of the model**,

$$z = w^T X + w_0$$

- But instead of predicting $y = z$ directly,
  we **apply a logistic function** to convert this linear combination into a **probability between 0 and 1**:

$$z = w^T X + w_0 \Rightarrow \hat{p} = \frac{1}{1 + e^{-z}}$$

- The function $\sigma(z) = \frac{1}{1+e^{-z}}$ is called the **sigmoid / logistic function**.

Linear Models and Regularization
1.2. Logistic Regression

- The output $\hat{p}$ represents the **probability of belonging to class 1**.
- Decision rule:

$$\text{Si } \hat{p} \geq 0.5 \Rightarrow \textit{classe 1} \; ; \; \textit{otherwise} \Rightarrow \textit{classe 0}$$

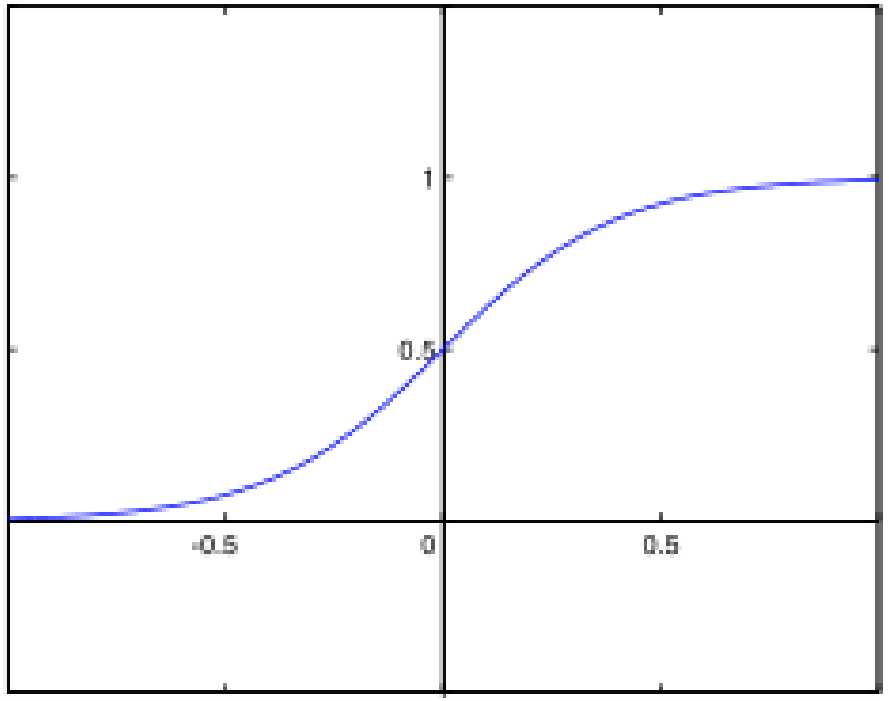
- In other words: logistic regression **retains the linear model**, but **changes the interpretation of the output**.
- Example:
  If $\hat{p} = 0.87$,the algorithm estimates a **probability of 87%** that the email is spam.

Linear Models and Regularization
1.2. Logistic Regression

- **The sigmoid function**
- The **sigmoid function** is a nonlinear S-shaped function:

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

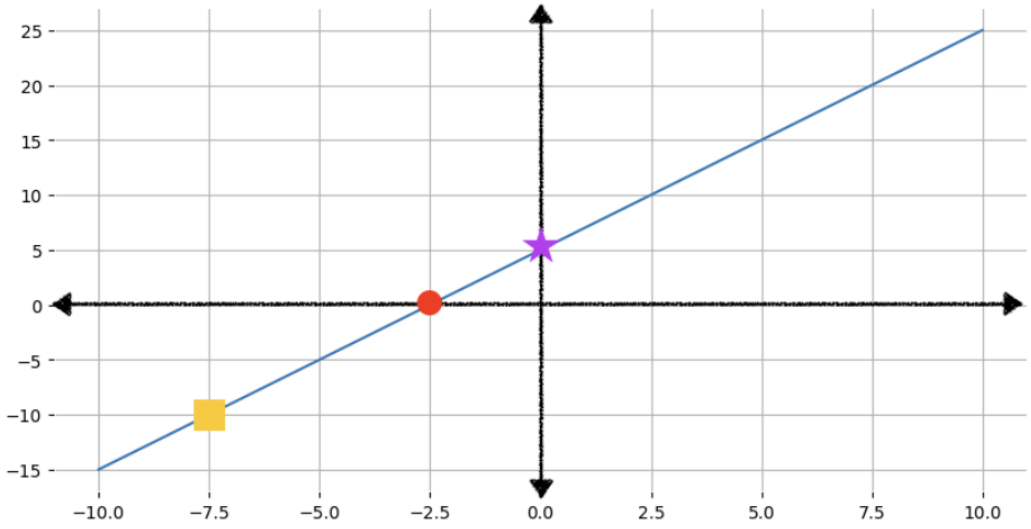
Property	Interpretation
$z \to -\infty$	$\sigma(z) \to 0$
$(z = 0)$	$\sigma(z) = 0.5$
$z \to +\infty$	$\sigma(z) \to 1$



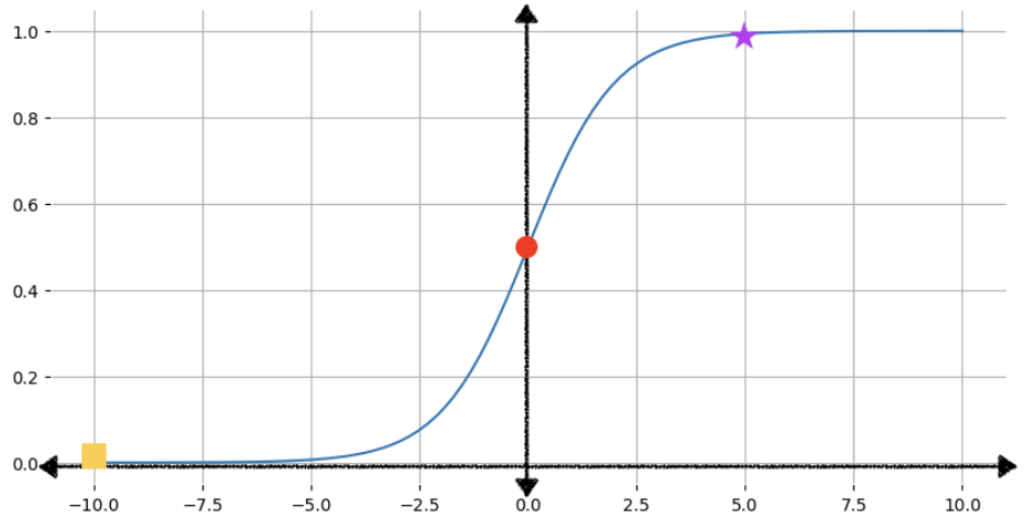
Linear Models and Regularization
1.2. Logistic Regression

- **The sigmoid function**
- The sigmoid “squeezes” any real value into the interval [0, 1].
- It maps the linear output to a probability of belonging to the positive class.

$z = 2x + 5$



$y' = 1 / (1 + e^{-z})$



Linear Models and Regularization
1.2. Logistic Regression

- **Cost function and model training**
- **The objective of training** is to find the **weights** w_j that minimize the prediction error between the estimated probabilities $\hat{p}$ and the true labels y.
- **Cost function used: the Log Loss (or cross-entropy)**

$$J(w) = -\frac{1}{m}\sum_{i=1}^{m}[y_i \log(\widehat{p_i}) + (1 - y_i)\log(1 - \widehat{p_i})]$$

- With:
  - y_i : True label (0 or 1).
  - $\widehat{p_i}$: Model prediction (a probability between 0 and 1).

Linear Models and Regularization
1.2. Logistic Regression

- **Cost function and model training**
- **Remarks**
- MSE is not suitable for classification:
  - it creates a non-convex cost function, making convergence difficult.
  - Several **local minima** and flat-gradient regions (“plateaus”) would be created:
    - the model could stop before reaching the optimal solution.

Linear Models and Regularization
1.2. Logistic Regression

- **Cost function and model training**
- **Remarks**
- The Logarithmic Loss (Log Loss) is the solution in this case
  - It is based on the **logarithm** of the magnitude of the variation, not on the simple distance.
  - it is convex (which guarantees that **Gradient Descent** will find the global minimum)
  - It heavily penalizes confidently wrong predictions (e.g., predicting 0.99 when the true class is 0).

Linear Models and Regularization
1.2. Logistic Regression

- **Cost function and model training**
- The minimization of $J(w)$ is performed by **gradient descent**, as in linear regression:

$$w_j \leftarrow w_j - \alpha \frac{\partial J(w)}{\partial w_j}$$

- with α = the learning rate.

Linear Models and Regularization
1.2. Logistic Regression

- **Worked example: predicting student success**
- We want to predict the **probability of passing an exam** based on the **number of study hours**.

Student	Study hours (x)	Pass (y)
1	0	0
2	10	0
3	20	0
4	40	1
5	60	1
6	80	1
7	100	1

- The objective is to find the parameters w_0 and w_1 that **maximize the likelihood** (or minimize the log-loss).

Linear Models and Regularization
1.2. Logistic Regression

- **Worked example: predicting student success**
- The cost function to be minimized is:

$$J(w_0, w_1) = -\frac{1}{m}\sum_{i=1}^{m}[y_i \log(\hat{p}_i) + (1 - y_i)\log(1 - \hat{p}_i)]$$

- where:

$$\hat{p}_i = \frac{1}{1 + e^{-(w_0 + w_1 x_i)}}$$

- In practice, a **gradient descent algorithm** or a numerical method (such as Newton-Raphson) is used to find w_0 and w_1.

Linear Models and Regularization
1.2. Logistic Regression

- **Worked example: predicting student success**
- After training the model on the data above, one often obtains values close to:

$$w_0 = -6 \; et \; w_1 = 0{,}2$$

- This gives:

$$z = -6 + 0{,}2x$$

- These values indicate:
  - When $x = 0$, the probability of success is very low ($z = -6 \Rightarrow p \approx 0{,}0025$).
  - As x increases, the probability grows progressively thanks to the slope $w_1 = 0{,}2$.

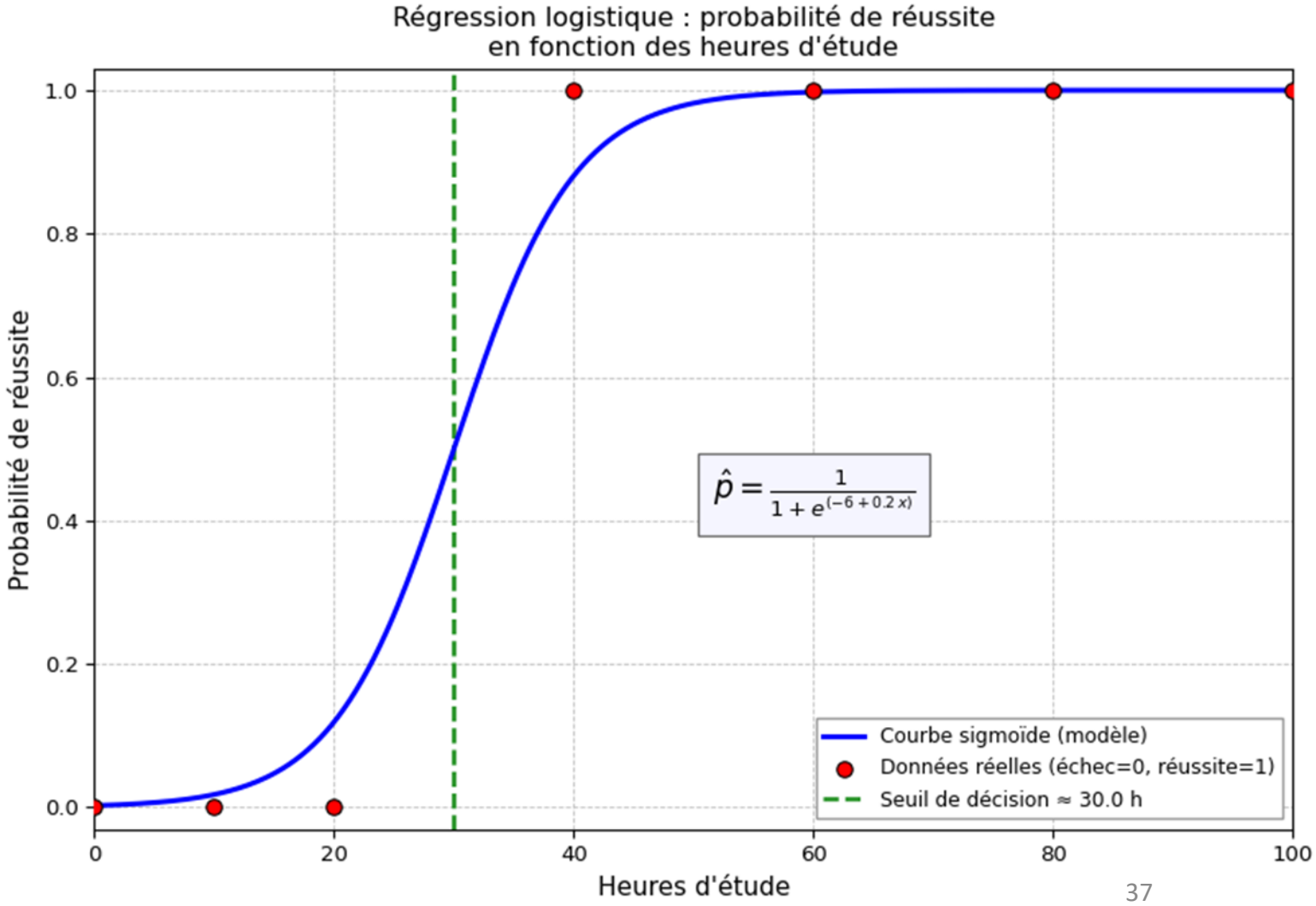
Linear Models and Regularization
1.2. Logistic Regression

- **Worked example: predicting student success**
  - **a) For $x = 0$:**
    - $z = -6 + 0.2(0) = -6$
    - $p = \frac{1}{1+e^{6}} \approx 0.0025$
  - **b) For $x = 20$:**
    - $z = -6 + 0.2(20) = -2$
    - $p = \frac{1}{1+e^{2}} \approx 0.12$
  - **c) For $x = 100$:**
    - $z = -6 + 0.2(100) = 14$
    - $p = \frac{1}{1+e^{-14}} \approx 0.9999992$

Linear Models and Regularization
1.2. Logistic Regression

- **Worked example: predi**

Student	Study hours (x)	Pass (y)
1	0	0
2	10	0
3	20	0
4	40	1
5	60	1
6	80	1
7	100	1

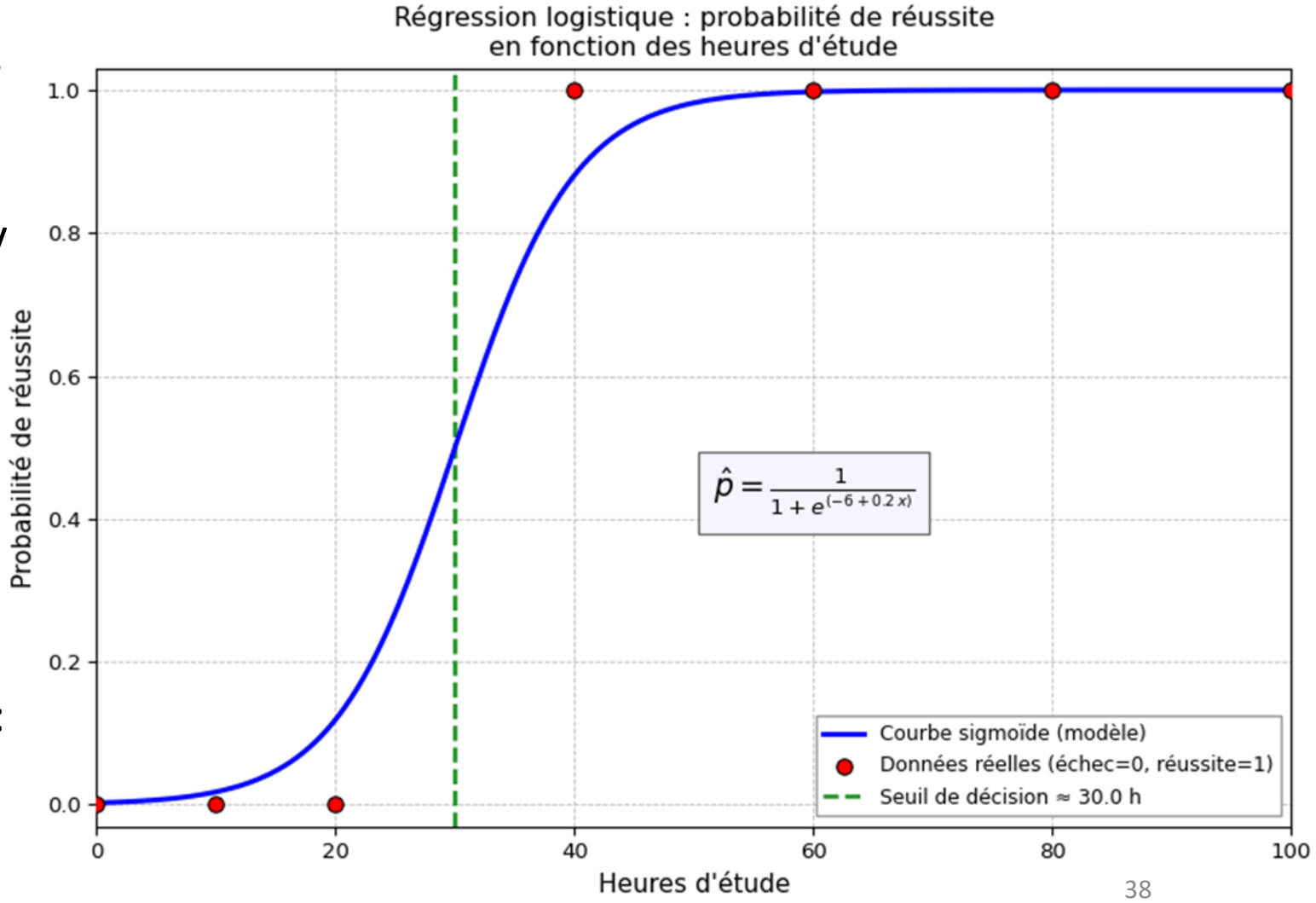


Linear Models and Regularization
1.2. Logistic Regression

- **Worked example: predi**

Interpretation

- The probability increases nonlinearly with study hours.
- The model estimates the decision boundary at 30 hours: beyond that point, the probability of success exceeds 0.5; it becomes almost guaranteed from 50 hours onward (p ≈ 0.98).
- The sigmoid function then saturates: beyond a certain threshold, the gains are negligible.



Linear Models and Regularization
1.2. Logistic Regression

- **Limitations of logistic regression**
- Linearity assumption
  - The model assumes that the relationship between the variables and the logit (log-odds) is linear. If the true relationship is nonlinear, the predictions will be biased.
- Performs poorly on data that is not linearly separable
  - If the classes strongly overlap in the variable space, a straight (or flat) boundary cannot separate them well.

Linear Models and Regularization
1.2. Logistic Regression

- **Limitations of logistic regression**
- Sensitive to correlated variables (multicollinearity)
  - Coefficients become unstable and difficult to interpret when two or more variables are strongly correlated.
- Does not handle complex interactions directly
  - The model does not naturally capture the combined effects of several variables; one must create **interaction terms** (X1×X2) or use **polynomial regression**.

Linear Models and Regularization
1.2. Logistic Regression

- **Comparison: Linear Regression vs. Logistic Regression**

Concept	Linear Regression	Logistic Regression
Problem type	Regression (continuous values)	Binary classification
Output function	$\hat{y} = w^T X + w_0$	$\hat{p} = \frac{1}{1 + e^{-z}}$
Output range	$]-\infty, +\infty[$	$[0, 1]$
Interpretation	Predicted value	Class probability
Cost function	MSE	Log Loss (cross-entropy)

Linear Models and Regularization
1.3. Regularization Techniques

- Linear and logistic regression adjust their parameters in order to minimize an error.
- In practice, these models may **learn the training data too well**, capturing noise rather than the general trend. This phenomenon is called **overfitting**.
- The solution is to introduce a **penalty term** into the cost function to prevent the coefficients w_j from taking excessive values.
  - This is called **regularization**

Linear Models and Regularization
1.3. Regularization Techniques

- **General principle of regularization**
- The cost function of a linear model becomes:

$$J(w) = Error(w) + \lambda \times Penality(w)$$

  - **Error(w)**: evaluates the performance of the model (MSE (for linear regression) or Log-Loss (for logistic regression)).
  - **Penalty(w)**: a term added to control the size of the weights (it controls model complexity).
  - λ **(lambda)**: regularization parameter (a hyperparameter) that controls the strength of the penalization (the trade-off between learning and constraint).

Linear Models and Regularization
1.3. Regularization Techniques

- **Effect of the regularization parameter λ**

λ	Effect	Risk
0	No regularization	Overfitting
Low	Slight constraint	Good trade-off
High	Coefficients near 0	Underfitting

Linear Models and Regularization
1.3. Regularization Techniques

- **The main types of regularization**

1. Ridge Regression (L2 Regularization)
2. Lasso Regression (L1 Regularization)
3. Elastic Net (L1 + L2)

Linear Models and Regularization
1.3. Regularization Techniques

1. **Ridge Regression (L2 Regularization)**

$$J_{Ridge}(w) = J(w) + \lambda \sum_{j=1}^{n} w_j^2$$

- The penalty is based on the sum of the squared coefficients (the bias w_0 is not penalized).
- Encourages the weights to be small, but never exactly zero.
- All features remain in the model, but their influence is reduced.
- **Use cases**
  - Very useful when variables are correlated (multicollinearity).
  - Stabilizes the coefficients → improves generalization.

Linear Models and Regularization
1.3. Regularization Techniques

1. **Lasso Regression (L1 Regularization)**

$$J_{Lasso}(w) = J(w) + \lambda \sum_{j=1}^{n} |w_j|$$

- Penalty based on the absolute value of the coefficients. Some coefficients are forced to exactly zero → automatic removal of marginally useful variables.
- **Use cases**
  - Ideal for datasets with a large number of variables (automatic feature selection).
  - Eases interpretation → simpler model.

Linear Models and Regularization
1.3. Regularization Techniques

1. **Elastic Net Regression (L1 + L2)**

$$J_{ElasticNet}(w) = J(w) + \lambda \left[\rho \sum_{j=1}^{n} |w_j| + (1-\rho) \sum_{j=1}^{n} w_j^2 \right]$$

- Combines the advantages of Lasso (selection) and Ridge (stability).
- ρ controls the proportion between L1 and L2.
- **Use cases**
- Recommended when:
  - the number of variables is large,
  - some are correlated,
  - a trade-off between simplicity and robustness is desired.

Linear Models and Regularization
1.3. Regularization Techniques

- Ridge (L2): penalizes coefficient magnitude → reduces variance, stabilizes.
- Lasso (L1): sets useless weights to zero → feature selection.
- Elastic Net: a combination of both, robust to correlations.